

Information Security Models and Algorithms Using Generative-Adversarial Neural Networks and User Data

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Abstract—In this article, the author reminds about the relevance of information security in the context of digitalization and active development of machine learning technologies. Also, the author emphasizes the popularity of users' use of weak passwords containing personal information about the user, which are vulnerable to hacking. To effectively identify such passwords, the author describes and proposes an approach to generating password lists based on a generative-adversarial neural network and known data about the user (such as full name, date of birth, hobbies, names of relatives, etc.), which can be used in information security audits to identify vulnerable passwords, penetration testing, as well as to solve some problems in forensic computer forensics. To train the neural network, it is proposed to use publicly available data from open sources that became known as a result of various personal data leaks, including compromised passwords, as well as information from various search engines and social networks such as «Vkontakte», «Odnoklassniki», and others.

Keywords—artificial intelligence, machine learning, PassGAN, generative-adversarial neural network, information security, password guessing, password auditing.

I. INTRODUCTION

The number of IS incidents and cyberattacks is growing every year. One of the reasons is compromised credentials. The methods of attacking credentials are continuously improving, especially during the period of rapid development of machine learning technologies.

Most attacks on information systems nowadays are related to the interception or selection of user passwords and their use. Thus, according to the information published by the information security company CyberArk - more than 80% of targeted attacks are aimed at breaking privileged accounts [1]. Similar results were published by the experts of Rostelecom-Solar Unsecure passwords were found in the systems and applications of the external perimeter of almost 80% of organizations [2].

Very often a user's password contains elements of personal information (full name, date of birth, hobbies, etc.), which greatly reduces password reliability. For example, the results of a survey conducted in 2022 by the Odnoklassniki project specialists together with the ResearchMe research center showed that almost every second respondent (49%) sets phone numbers, initials, and birth dates in their passwords [3].

The problem is that identifying weak passwords during information security audits is time-consuming, and the effectiveness is often insufficient. At the same time, the audit process often uses standard, well-known password dictionaries that do not take into account personal information of employees. Thus, this leads to insufficient audit efficiency because potential passwords based on personal data are not

detected and excluded[4, 5]. This fact negatively affects the number of successful cyber-attacks in the context of social engineering methods, as well as the use of intelligent models based on neural networks [6, 7].

To identify such insecure passwords, the project proposes to implement an approach that combines a password generation method based on a generative-adversarial neural network and trained on an array of real passwords from published leaked information together with enriched data about the owners of these passwords in order to generate highly reliable hypotheses about password usage.

The problem to be solved is to develop a model to improve the efficiency of the process of identifying weak passwords consisting of personal user data by developing a tool for generating lists of personalized passwords using a generative-adversarial neural network.

II. EXISTING PASSWORD VERIFICATION METHODS AND AN ALTERNATIVE

Various password auditing methods are used to detect weaknesses in password security, which most often consists in identifying "weak" passwords in advance and neutralizing them (e.g., HashCat [8], John the Ripper [9], Hydra [10], Medusa [11], CUPP [12], RainbowCrack [13], etc.). The audit evaluates the strength of passwords depending on its length, the strength of the alphabet used, its presence in published leaks, and other characteristics that affect the difficulty of cracking passwords[4].

Existing tools allow generating password dictionaries given a set of specified user data, but this tool is based on the use of predefined patterns that are combined with each other in a certain way without using machine learning technologies [12].

In order to identify such insecure passwords, the project implements an approach that combines a password generation method based on a generative-adversarial neural network and trained on an array of real passwords from published leaked information together with enriched data about the owners of these passwords in order to generate highly reliable hypotheses about password usage.

The services "Dehashed"[14], "God's Eye" [15], social networks [16, 17], other publicly available services, and sources of information are used for data collection and enrichment [15], social networks [16, 17], other publicly available services, and information sources. "Fused" databases contain a huge amount of information. If we take into account the fact that most users in the network voluntarily leave a lot of information about themselves, for example, in social networks or on all sorts of sites, the aggregate of such information can be a real threat in the hands of an attacker.

Having a minimum of information about a person, it is possible to determine his full name, date of birth, get information about his relatives, place of residence and work, availability of a car, hobbies, passport data and even passwords that he has ever used.

From the point of view of the legislation, information posted by a user on the Internet (public profile in a social network, on websites, forums and other online resources) is publicly available information. This is stated in part 1 of article 7 of the Federal Law of July 27, 2006 № 149-FZ "On Information, Information Technologies and Information Protection" "Publicly available information includes publicly known information and other information, access to which is not restricted". Of course, this concept also includes information posted in the public domain on the Internet [18]. In accordance with part 2, article 10.1 of the Federal Law of 27.07.2006 No. 152-FZ "On personal data" It is not necessary to request a separate and written consent for the processing of publicly available personal data [19]. Roskomnadzor itself clarified on its official website in which cases the consent of the subject of personal data is not required for the processing of personal data: when personal data is processed, access to which is granted by the subject of personal data or at his/her request [20].

At the same time, it should be taken into account that the processing of personal data authorized by the personal data subject for dissemination should be stopped at any time at the request of the personal data subject [19].

That said, this approach should show good results because recurrent and similar neural networks can perfectly combine learned elements from training data, but they are less suitable for creating new passwords consisting of elements not included in the training dataset. At the same time, generative adversarial networks (GANs), which have a generator that uses a vector with random input noise, can produce new and unique results while still being able to take into account the user input. Thus, GANs have the ability to generate more diverse and original passwords than recurrent neural networks [21].

III. THE PROPOSED METHOD BASED ON GAN NETWORK

The logic of operation of generative-adversarial networks within the framework of the considered problem can be represented in the form of the scheme shown in Fig. 1.

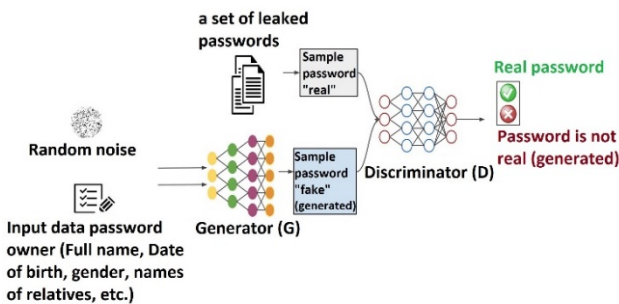


Fig. 1. Model of a generative-adversarial neural network.

According to the logic of the generative-adversarial network, the intelligent model receives as a training sample data on password owners who have been assigned a corresponding, previously used password (Fig. 2; personal

information is hidden using blur). In the process of training, the machine learning model aggregates the most specific features and patterns of each user from the training sample.

Login	Password	URL	Name	Surname	Date of birth	Maiden name	Partner's name
790	1967AHHA	vk.com/id	ANNA		09.04.2022	Burkova	
798	twomkey21	vk.com/id	Twomkey		06.06.2001		
795	polinashka909	vk.com/id	Polina		26.02.2006		
796	fedotova30092003	vk.com/id	Fedotova		30.09.2003		
792	457515Radik	vk.com/id	Radik		10.02.1997		
795	nastya35	vk.com/id	Anastasiya		22.12.1991		
795	ValeriaBobrov1998	vk.com/id	Valeria		26.07.1998	Bobrovnikova	
791	fomina1	vk.com/id	Tatiana		27.10.2022	Fomina	Nikita
791	лерамолякена	vk.com/id	Artem		02.11.2022		Valeria
790	Nadezhda.09.06.92	vk.com/id	Nadezhda		09.06.1992	Zolina	
797	25031983AS	vk.com/id	Anna		25.03.1983		
790	1412AlenoChka	vk.com/id	Alena		14.12.2000		

Fig. 2. A fragment of a training sample containing data on password owners from open sources.

GAN is a teacherless machine learning algorithm based on a combination of two neural networks, one of which generates samples and the other tries to distinguish correct ("genuine") samples from incorrect ones [21].

GAN consists of two neural networks, a discriminator and a generator. The generative model can be thought of as analogous to a team of counterfeiters trying to make counterfeit currency and use it without detection, and the discriminator model can be thought of as the police trying to detect counterfeit currency. The competition in this game encourages both teams to improve their methods until the fakes are indistinguishable from the genuine article.

It is easiest to apply the adversarial modeling scheme when both models are multilayer perceptrons. To learn the distribution of the generator P_g over the data x , a prior on the input noise variables $P_z(Z)$, is defined, then the mapping to the data space is represented as $G(z; \theta_g)$, where G is a differentiable function represented by a multilayer perceptron with parameters θ_g . A second multilayer perceptron $D(x; \theta_d)$, which produces a single scalar, is also defined. $D(x)$ represents the probability that x is obtained from the data rather than from P_g . We train D to maximize the probability of assigning the correct label to both training samples and samples from G . Simultaneously train G to minimize $\log(1 - D(G(z)))$:

In other words, D and G play the following two-player minimax game with a value function $V(G; D)$, this is represented below in Equation 1 [18]:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

The advantage of this method is that an algorithm trained on a set of real user data will allow us to approach password generation logic more similar to human thinking. During the training process, the machine learning model aggregates the most specific features and patterns of each user from the training sample. In the vast majority of cases, passwords will be generated rather than recovered from the training sample, since the same password can be generated by the intelligent system in different variants, depending on the given input data. The generator, which is a part of the intelligent system, provides generation of such a password, which will most closely match the input data received from the user. This will allow for more efficient detection of weak passwords, which are more related to the user's personal data.

Timely auditing and elimination of such passwords will allow to detect weaknesses that can lead to information security breaches. An example of using such a tool to solve the password auditing problem in an organization is shown in Fig. 3.

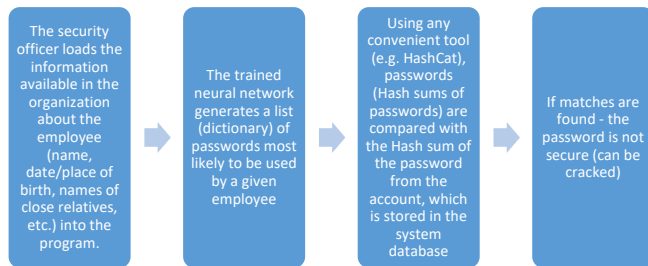


Fig. 3. Example of using the tool for an audit task.

The result of the work will be software (trained intelligent model) allowing to generate a list-dictionary of possible user passwords, based on known personal data of the user + analysis of the efficiency of using this approach and comparison with traditional solutions.

Such a tool can be used to conduct various information security audits, penetration testing (pen-test), and can detect weaknesses that may lead to information security breaches. In addition, the tool can be effective for forensic computer forensic examination to detect and analyze forensically significant information. An example of a fragment of the password list generated using the proposed algorithm is shown in Fig. 4.

01.02.2003-Mihaillosev	188losev
Mihaillosev	188losev188
Mihaillosev0	losev188
@Mihaillosev	188losev
@Mihaillosev0	188losev188
Mihaillosev2003	misha188
2003Mihaillosev	188misha
Mihaillosev01022003	188misha188
Mihaillosev01.02.2003	Misha188
01022003Mihaillosev	188Misha
01.02.2003Mihaillosev	188Misha188
Mihaillosev_01022003	olga188
Mihaillosev_01.02.2003	188olga
01022003_Mihaillosev	188olga188
01.02.2003_Mihaillosev	Olga188
Mihaillosev_01022003	188Olga
Mihaillosev-01.02.2003	188Olga188

Fig. 4. An example fragment of the generated password list using the algorithm.

Since statistics shows that most users use passwords related to their personal lives, this approach will increase the efficiency of password auditing and contribute to improving information security.

CONCLUSION

In conclusion, the growing prevalence of IS incidents and cyberattacks underscores the critical need for strong password protection measures. Compromised credentials remain a frequent cause of security breaches, with attackers often targeting privileged accounts and using weak password methods. Traditional password verification methods, while valuable, often fail to effectively detect weak passwords, especially those derived from user identities.

The proposed approach of utilizing generative-adversarial neural networks trained on augmented password data offers a promising solution to the widespread problem of weak password security. By integrating machine learning techniques, this method improves the identification of passwords derived from personal information, thereby enhancing the defense against cyber threats. Its potential goes beyond password auditing to include penetration testing and forensic computer forensics. Utilizing such innovative approaches is critical to protecting sensitive information and improving overall cybersecurity resilience.

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